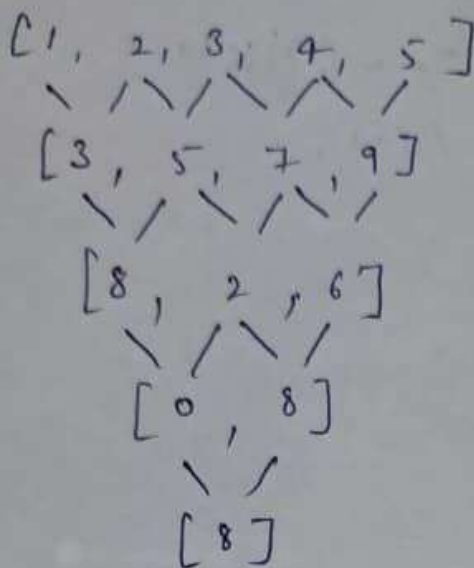


(Experiment 12)



So, the $\text{Num}[i] = (\text{num}[i] + \text{num}[i+1]) \% 10$

we have to continue until only one element remains.
& then return the element.

- reduce the array step by step.
- At each step
replace each element with the sum of adjacent $\% 10$.
- continue until one element left.

Algo 1 $n = \text{size of nums.}$

while ($n > 1$):

for $i = 0$ to $n - 2$:

$\text{num}[i] = (\text{num}[i] + \text{num}[i+1]) \% 10$

reduce size $n = n - 1$

return $\text{nums}[0]$

Time complexity = n^2

space = 1

optimal & Based on Pascal's Triangle

result = (result + (comb * num[i]) % 10
comb = comb * (n - i) / (i + 1);

}

~~return res~~

code &

```
int n = num.size();
```

```
int res = 0;
```

```
int comb = 1;
```

```
for (int i = 0; i < n; i++) {
```

```
    res = (res + comb * num[i]) % 10;
```

```
    comb = comb * (n - 1 - i) / (i + 1);
```

```
}
```

```
return res;
```

```
}
```

time complexity = $O(n)$

space complexity = $O(1)$